Glory Aditya Jauzaulbahi

gloryadityaj@gmail.com | +62-858-9537-7701
linkedin.com/in/gloryadityaj | github.com/adityaglory

PROFESSIONAL SUMMARY

Engineering graduate in Biomedical Engineering specializing in Computer Vision, Generative AI, model deployment & MLOps. Possesses a foundation and practical background in building end-to-end model pipelines, from dataset curation and preprocessing into model deployment and evaluations, bridging gap between raw data and visual perception. Committed to applying mathematical models and engineering method to solve complex data flows and engineering challenges.

TECHNICAL SKILLS

Languages: Python, SQL.
AI & ML Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, LangChain.
Computer Vision: OpenCV, MediaPipe, Roboflow.
MLOps & Infrastructure: Docker, MLflow, Streamlit, GitHub Actions.
Tools & Cloud: Git/GitHub, Linux (Ubuntu/Raspbian), Microsoft Azure AI.

EDUCATION

Universitas Gadjah Mada — Yogyakarta, Indonesia
Bachelor of Engineering in Biomedical Engineering — *Graduate: Aug 2025*

- **Focus:** Biomedical Signal & Image Processing, Differential Equations, Linear Algebra, Biomedical Artificial Intelligence, Intelligence Systems.

Pacmann AI — Online
AI/ML Engineering Bootcamp — *Jan 2024 - March 2026*

- **Focus:** Data Visualization & Analysis, Probability & Statistics, Recommender System, MLOps, Natural Language Processing.

PROFESSIONAL EXPERIENCE

Pusat Riset Sains Data & Informasi, BRIN — Bandung, Indonesia
AI Research Intern — *Jul 2022 - Sep 2022*

- Engineered a computer vision pipeline utilizing MediaPipe to extract temporal features from collected processed video dataset, attaining an average detection accuracy of 79% with SVM model.
- Conducted device testing across 6 varying hand gesture in 2 situations, achieving real-time hand motion classification at 1-4 FPS on Raspberry Pi 4.
- Documented research and experiment results from dataset of over 6000 hand-wash gesture image sequences for internal knowledge transfer.

Project Repository

PROJECTS EXPERIENCE

Meltion (Melanoma Detection App) - Capstone Team Project

- **Role:** AI Engineer.
- Preprocessed dataset of $\pm$ 10000 diverse skin lesion dataset, by engineering a robust preprocessing workflow for illumination correction and data augmentations.
- Fine-tuned a pretrained ResNet50 model to optimize accuracy within dataset, achieving 78% accuracy in multiclass detection while maintain high recall value through data balancing strategies.

- Collaborated closely with the mobile developers to integrate pre-trained model and backend via Google Vertex AI.

Data Augmentation for Imbalanced Dataset with GANs - Undergraduate Thesis

- Conducted novel research on addressing severe class imbalance in medical datasets by architecting custom GANs and VAEs based on previous approach.
- Synthesized over 100 high resolution medical data using GANs and VAE architectures to directly address dataset scarcity and class imbalance.
- Benchmarked the perceptual realism of synthetic outputs against ground-truth data distributions, achieving an optimal Fréchet Inception Distance (FID) score of 0.2249 for GANs model.
- Benchmarked the synthetic data's utility by retraining classifiers, proving a significant improvement in F1-score (96%) compared to baseline traditional augmentation methods.

Project Repository

Biomedical Research Local RAG System

- Investigated and implemented a RAG architecture using Llama 3.2 to extract insights from unstructured biomedical papers, with utilize Maximal Marginal Relevance (MMR) and optimizing context window parameters ($k = 12$) for context aware querying.
- Engineered a scalable pipeline using Docker and ChromaDB, reducing information retrieval latency speeds while maintaining semantic accuracy.
- Implemented document embedding and vector search methodologies, reducing information retrieval time to under 2s seconds with high semantic accuracy.

Project Repository

End-to-End Nutritional OCR & Data Extraction Pipeline

- Architected a Computer Vision pipeline utilizing a fine-tuned YOLO model and EasyOCR, reducing manual data entry time while maintaining high accuracy on noisy, high glare packaging.
- Engineered a processing engine with fuzzy Regex matching, successfully recovering 100% of missing packaging metadata via automated cross box imputation and validating macronutrients using the Atwater system.
- Developed a highly scalable dual interface system (Streamlit Web UI & CLI) capable of batch processing 40+ images seamlessly, implementing strict distance bounding logic to eliminate OCR hallucination errors.

Project Repository

CERTIFICATIONS

Azure AI Fundamentals (AI-900) *Microsoft, Sep 2025*

- Validated understanding of computer vision workloads and cloud-based AI deployment strategies.

DeepLearning.AI GANs Specialization *Coursera, Aug 2023*

- Advanced specialization in generative modeling, focusing on feature representation and image reconstruction techniques.

LEADERSHIP EXPERIENCE

UKM Perisai Diri, Universitas Gadjah Mada 2020 - 2021

- **Head of Internal Division (2021):** Directed organizational workflows and optimized internal resource allocation.
- **Inventory Division Supervisor (2020):** Managed asset tracking systems and resource accountability for organizational activities.